CHAPTER

29

Dentin Hypersensitivity

With increase in life expectancy and increased retention of natural dentition, the incidence of dentin hypersensitivity is likely to be a cause of concern for the dental professionals. Dentin hypersensitivity is an exaggerated response of exposed dentin to external stimuli, regardless of its location.

Dentin hypersensitivity is characterized by sharp and short pain, which arises from exposed dentin in response to a variety of stimuli, viz. thermal, evaporative, tactile, osmotic or chemical; the pain cannot be ascribed to any other form of dental defect/pathology. Another term, root sensitivity, has also been suggested for dentin hypersensitivity arising from gingival recession in periodontal disease and following periodontal treatment. This group of patients may have micro-organisms invading the root dentinal tubules, coinciding with recession in these areas. This condition may be of different etiology, but results in similar pain symptoms.

INCIDENCE AND PREVALENCE

Dentin hypersensitivity is mostly prevalent between 20 and 50 years of age, with peak incidence between 30 and 40 years. Later in life, age-related changes in pulpal processes may result in a reduction of sensitivity, owing to reparative processes (secondary and tertiary dentin), fibrosis in the pulp and sclerosis of the tubules, which will decrease permeability and reduce the hydraulic conductance of dentin. Women are more frequently affected, may be because of better oral hygiene and dietary practices. The reported prevalence of dentin hypersensitivity varies from 3 to 67%; however, the most consistent figure documented is 15%.

Dental hypersensitivity after periodontal treatment has reported higher prevalence values, in the range of 60–90%.

Regarding the type of teeth involved, canines and premolars of both the arches are the most affected teeth, especially their buccal-cervical areas.

ETIOPATHOGENESIS

Dentin is a vital tissue, consisting of dentinal tubules along with extensions of odontoblasts, as part of dentin–pulp complex. The odontoblastic processes are surrounded by dentinal fluid inside the dentinal tubules. The fluid around 22% of the total volume is an ultrafiltrate of blood from the pulp, containing organic proteins and inorganic salts.

The dentin hypersensitivity occurs when the external stimulus contacts exposed dentin, triggers a rapid outflow of dentinal fluid, and the resultant pressure change activates interdental nerve fibers, causing immediate pain. Tactile, cold and osmotic stimuli, all trigger rapid fluid outflow. Heat, on the other hand, triggers a slow retreat of dentin fluid and the resultant pressure change activates the nerve fibers slowly; cold is generally more problematic than heat.

In case, the protective covering of tooth (enamel/cementum) is lost, may be because of attrition, abrasion, erosion or periodontal therapies; the smear layer, if present over the eroded surface, may prevent initiation/localization of dentin hypersensitivity, especially in cervical areas.

A number of theories have been proposed over the years to explain the pain mechanism of dentinal hypersensitivity.

1. Direct Neural Stimulation

An early hypothesis suggests that dentin hypersensitivity is caused by the direct stimulation of sensory nerve endings in dentin. On the basis of microscopic and experimental data, it seems unlikely that neural cells exist in the sensory portion of the outer dentin. This theory is not accepted.

2. Odontoblast Receptor Theory

The odontoblast receptor theory suggests that odontoblasts act as receptor cells, mediating changes in the membrane potential of the odontoblasts via

synaptic junctions with nerves. This could result in sensation of pain from the nerve endings located in the pulpodentinal border. However, odontoblast cells are not considered excitable cells and no synaptic junction could be demonstrated between odontoblasts and nerve terminals. The theory is inconclusive.

3. Hydrodynamic Theory

The hydrodynamic theory, proposed by Brannstrom, states that dentin hypersensitivity may be caused by movement of the dentinal tubule contents. An increased inward/outward fluid flow causes a pressure change, leading to distortion of the nerve fibers by a mechanoreceptor action, inciting sharp, shooting pain (Fig. 29.1A, B).

The width of the tubule is very important, as the rate of fluid flow is dependent on the fourth power of the radius. If the tubule diameter doubles, a 16-fold increase in fluid flow results. Sensitive teeth have wider (2 times) tubules at the buccal-cervical area as compared to non-sensitive teeth. A higher velocity of fluid flow in tubules of smaller diameter may provoke pain sensations. Dentin will only be sensitive, if the tubules are patent from the pulp to the oral environment.

The trigeminal nerve supplies the pulp, with innervations from myelinated fibers (Aα and Aδ) and non-myelinated C fibers. It is proposed that the larger myelinated fibers (Aα and Aδ) can respond to stimuli that displace the fluid in the dentinal tubule through a hydrodynamic mechanism, such as tactile, evaporative, osmotic or thermal challenges, to elicit short, sharp, stabbing pain that lasts for only a few seconds.

Subsequently, the pain experienced with dentin hypersensitivity can persist as a dull, throbbing ache for variable periods of time (Both types of pain, sharp/short and dull/long, prevalent in exposed dentin).

The status of the pulp in dentin hypersensitivity is not known; may be minor inflammation as a result of length of time the symptoms persist. It is hypothesized that when the pain continues as a throbbing ache, pulpitis can be present.

Dentin hypersensitivity results as the dentin is exposed from either coronal or radicular regions of the tooth.

a. *Exposure of Dentin due to Loss of Cementum*

Gingival recession is a common finding both in populations with high standards of oral hygiene as well as in populations with poor oral hygiene. The former type of recession can be attributed to overzealous tooth brushing and is predominantly found on the buccal surfaces of canines and premolars. Teeth and tooth surfaces that receive most brushing during the brushing cycle overall show the highest predilection for recession.

The latter type of recession is seen in periodontal patients, during and following the treatment. These lesions can be found anywhere around the root. Dentin hypersensitivity can be attributed to both types of recession. The two etiologies result in the same pain condition; however, they should be treated differently, especially in terms of prevention.

b. *Exposure of Dentin due to Loss of Enamel*

Enamel is highly susceptible to acid erosion, which is probably the most aggressive type of wear compared with abrasion and attrition. The acid may be derived from intrinsic, gastric, or more often, extrinsic sources that are dietary in origin. If the erosive challenge is followed with the abrasive challenge, the softened

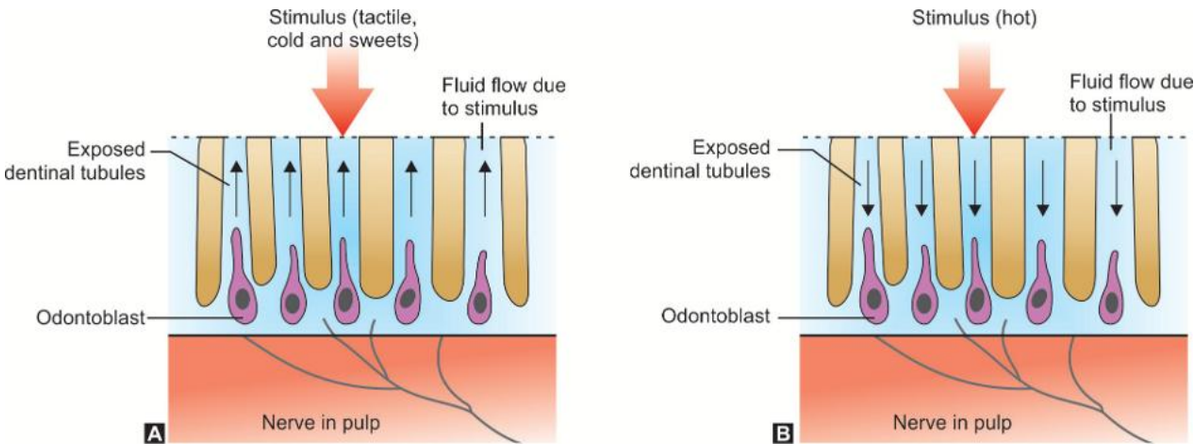


Fig. 29.1: Hydrodynamic theory: **(A)** Outward flow of dentinal fluid; **(B)** Inward flow of dentinal fluid

hard tissue will be easily removed. One should wait for at least half an hour for tooth brushing after having an acidic drink.

Toothbrushes alone have no measurable effects on enamel and dentin loss. But abrasive toothpastes accomplished with overzealous tooth brushing can lead to enamel loss. Attrition of tooth surfaces is usually the result of parafunctional activity.

FACTORS AFFECTING HYPERSENSITIVITY

- Enamel provides greater protection of the underlying dentinal tubules; also provides electrical resistance.
- Cementum provides less protection. Loss of the cementum easily occurs with use of a hard toothbrush and/or an abrasive toothpaste, or by root scaling and planing during periodontal therapy.
- Salivary calcium and phosphate ions facilitate obturation of the dentinal tubules, affecting their permeability.
- State of the pulp may affect the sensitivity (Inflamed pulpal tissue may result in greater sensitivity than normal; whereas, necrotic pulp tissue results in lower sensitivity).
- Longevity of the stimulus is an important feature (Time required for the pulp tissues to return to baseline values affects the tooth sensitivity).

MEASURING TOOTH HYPERSENSITIVITY

The methods used to measure tooth hypersensitivity are as follows:

a. Tactile

The tactile method implies lightly passing a sharp dental explorer over the sensitive area of a tooth (usually along the cementoenamel junction). The patient's response is graded on a severity scale, generally 0 to 3 (a score of '0' is assigned if no pain is felt, '1' if there is slight pain or discomfort, '2' if there is severe pain, and '3' if there is severe pain that lasts for more than 10 seconds.

The force-sensitive electronic probe, used to measure the depth of periodontal pockets at fixed pressures, is also used to quantify the sensitivity. The operator can vary the force applied to the tip by regulating the amount of current and the tip position.

The probe force is controlled within ±1 gram. The probe force can be increased in steps (say 5 gm) until the subject experiences discomfort. That point is taken as the pain threshold. If a maximum force of 70 grams is reached with no discomfort, the tooth is scored as non-sensitive.

b. Thermal

Room air is used to test the sensitivity since it is cooler than the oral temperature. The patient will feel pain when cool air contacts the sensitive tooth. Blowing air on a tooth leads to drying the area, which can also be stimulatory.

Tactile stimuli are applied before thermal stimuli, if the two are being used in the same subject.

Testing with heat is carried out keeping the basic temperature at 37.5°C and increasing the temperature in increments of 1°C to the point at which pain is felt.

c. Osmotic

After isolation of the test tooth, a cotton applicator saturated with the sucrose solution is applied to the root surface of the tooth and allowed to remain in place for 10 seconds. The sensation as *no pain* (designated as '0') or *pain* (designated as '1') is recorded. The tooth surface is later rinsed with warm water.

d. Electrical

Electrical pulp testers are widely used to determine the vitality of tooth. The quantification of electric current can be used for measuring tooth sensitivity. Certain pulp testers may determine a condition known as pre-pain. The pre-pain sensation has been attributed to the larger, more rapidly conducting nerve fibers than the smaller diameter nerve fibers.

For safety reasons, any electrical current that is applied should be limited to less than 1.0 milliampere, preferably 0.5 milliampere (the approximate range of the human body's threshold of current perception). The voltage pulses with a width of 0.05 to 0.20 millisecond each and spaced at 5 to 10 millisecond intervals will provide a stimulus of pre-pain instead of pain.

e. Air Indexing

A restricted air stream is directed toward the cemento-enamel junction at approximately 45° angle for 0.5 to 1.0 second. The tip is kept at a distance of 0.5 cm from the tooth cervix. The duration of the air exposure is for shorter duration to avoid neurogenic inflammation or evaporative phenomenon. The air is applied through block hole (Different holes have different pressures, viz. No. 1: 2–3 psi, No. 2: 4–6 psi, No. 3: 11–17 psi, No. 4: 25–30 psi and No. 5: 35–40 psi). Usually number one hole is used to initiate the response. If no response is elicited, then the next number is used. The least air pressure necessary for detection of sensitive tooth is termed as 'threshold patient response'. These threshold response values are recorded as none (0), slight (1), moderate (2) or severe (3).

The same method is used for lingual surfaces. Resultant threshold response can be noted. The changes in response over time is evaluated before planning treatment for dentinal sensitivity.

DIAGNOSIS

The following features are important for diagnosing dentin hypersensitivity:

i. Diagnosis of dentin hypersensitivity should be based upon history and clinical examination.
ii. Differential diagnosis, to exclude other conditions giving rise to similar pain symptoms.
iii. Identification of etiologic and predisposing factors, particularly dietary and oral hygiene habits, pertinent to erosion and abrasion.
iv. Removal or minimizing etiologic and predisposing factors through dietary advice and oral hygiene instructions.
v. Recommendation of treatment should be based upon individual needs.
vi. Treatment of all secondary condition with symptoms similar to dentin hypersensitivity.

Differential Diagnosis

Prior to advocating treatment regimen, it is important to exclude the conditions simulating symptoms of dentin hypersensitivity (differential diagnosis). A number of dental conditions can give rise to pain symptoms similar to those of dentin hypersensitivity. Indeed, a definitive diagnosis of dentin hypersensitivity is reached through exclusion of the following conditions, which may need different treatment options.

- Cracked tooth syndrome, often in heavily restored teeth.
- Incorrect placement of dentin adhesives leading to nanoleakage.
- Fractured restorations and incorrectly placed dentin pins.
- Pulpal response to caries and to restorative treatment.
- Inappropriate application of medicaments during cavity preparation.
- Restorations in traumatic occlusion.
- Palatogingival groove and other enamel invaginations.
- Chipped teeth, exposing dentin.
- Vital bleaching.

Table 29.1 summarizes differential diagnosis of pain due to dentin hypersensitivity.

Table 29.1: Differential diagnosis of dental pain during dentin hypersensitivity

Conditions simulating pain of hypersensitivity	*Characteristic features*
Caries	As caries penetrates dentinoenamel junction and invade dentin, patient feels sensitivity: Clinical examination is sufficient.
Cracked tooth syndrome	Sharp intermittent pain on biting and relief of pain as the pressure is withdrawn: Bite test (a tooth sleuth) can be used to evaluate pain.
Fractured tooth	Fracture involving dentin induces sensitivity/pain; clinical examination includes:
Pulpitis	• Sharp pain provoked by hot, cold, or sweet; disappears after removal of stimulus (reversible pulpitis). • Sharp, throbbing, intermittent pain provoked by hot, chewing, lying flat; persists even after removal of stimulus (irreversible pulpitis).
Periapical periodontitis	Continuous dull pain, increased on biting.
Tooth substance loss	Pain and sensitivity to heat and cold due to wearing of the tooth surface, leading to dentin exposure.
Postoperative sensitivity	• Sensitivity due to cavity preparation – Heat generation (lack of adequate cooling during cutting of tooth structure) – Exerting excessive pressure – Bur eccentricity (abnormal vibrations) • Sensitivity due to restorative phase (postoperative) – Leakage – Improper bonding procedure – Abnormal cuspal stresses – Improper cementation of indirect restorations

MANAGEMENT OF HYPERSENSITIVITY

The following features are important in management of hypersensitive dentin before initiating specific treatment modalities.

1. Dietary Counseling

Dietary acids are capable of causing erosive loss of tooth structure, subsequently opening the dentinal tubules. Dietary counseling should focus on the quantity and frequency of acid intake. It is established that wine, citrus fruit juices, apple juice, and yogurt were capable of dissolving the smear layer; however, tannic acids, low-pH carbonated drinks and Coca-Cola had no effect on smear layer. Loss of enamel is increased when brushing is performed immediately after exposure of the tooth to dietary acids.

2. Tooth Brushing Techniques

Incorrect tooth brushing, especially using hard bristles, may lead to abrasion. Instruction of proper brushing techniques can prevent loss of enamel and the resulting hypersensitivity.

3. Plaque Control

Saliva contains calcium and phosphate ions and is able to contribute to the formation of mineral deposits within exposed dentinal tubules. The presence of plaque may interfere with this process, as plaque bacteria produce acid and dissolve any mineral precipitates, thus opening dentinal tubules.

Effective oral hygiene measures should be followed.

TREATMENT MODALITIES

Various desensitizing agents (Fig. 29.2) have been used and are currently being tried to treat dentin sensitivity. Dentin fluid flow rate is proportional to the fourth power of the radius of dentinal tubules; certainly of clinical relevance to the treatment of dentin hypersensitivity.



Fig. 29.2: Desensitizing agents

The requisites of desensitizing agents are:

- Should provide immediate and lasting relief of pain
- Easy to apply
- Well tolerated by patients
- Should not be injurious to pulp
- Should not discolor the tooth
- Relatively inexpensive

The treatment procedures are categorized as 'at home' desensitizing therapy and 'In-office' treatment modalities.

A. 'At-home' Desensitizing Therapy

A dentifrice is defined as an agent, usually incorporated in toothpastes, helps in cleaning the accessible surfaces of the teeth. The agent containing desensitizing formulation in a dentifrice can be used for alleviating symptoms of tooth sensitivity.

Dentifrices usually contain abrasive, surfactant and thickener, etc. Therapeutic dentifrices contain medicament in addition to the above mentioned ingredients.

Abrasives are solid particles that clean or polish the tooth surface; for example, calcium phosphate, pyrophosphate, sodium metaphosphate and alumina. Silica is preferred because of its compatibility with fluoride. Dentifrice abrasives are essential to prevent tooth staining.

The taste of a dentifrice is one of the most important factors related to the continuous use of a particular dentifrice.

Patients using desensitizing pastes should be advised to use minimal amount of water to prevent the dilution of the active agent. Desensitizing mouth-washes and chewing gums can reinforce the action of toothpastes. The 'At-home' therapy must be reviewed periodically to assess the outcome; otherwise, patient is referred for 'In-office' treatment.

Toothpastes are effective, but take longer time to achieve pain relief (average 4–8 weeks). Another challenge with use of toothpaste is compliance, and difficulty of delivery to specific sites, which may encourage non-compliance.

The hypothesized mechanisms of action of desensitizing agents are:

i. Blocking dentinal fluid movement by occluding dentinal tubules
ii. Blocking pulpal nerves activity by altering the excitability of sensory nerves.

Various agents used as desensitizing agents are as follows:

a. *Strontium Chloride*

Strontium chloride (10%) was the first tubule blocking ingredient to be introduced into toothpaste. The three

potential mechanisms of action for strontium salts can be:

i. Depolarization of nerve
ii. Strontium may replace lost calcium in the hydroxyapatite lattice to strengthen demineralized enamel and/or dentin.
iii. Strontium salts may deposit a layer of fine particles to occlude tubules. The third has been proposed to be the most likely of the three mechanisms. However, it has been established that strontium salts have minimal effect in reducing the symptoms of dentin hypersensitivity.

b. *Potassium Salts*

Potassium salts are the most commonly used agents in toothpastes. The mechanism of action of potassium salts is that they can reduce the excitability of intradental nerves. Raising the concentration of potassium ion above the physiological level in the extracellular fluid induces depolarization of the nerve cells, following which the nerves become unresponsive to excitatory stimuli (Calcium and strontium are also able to suppress nerve activity, but to a much lesser extent than potassium). The potassium ion has to diffuse into the dentin tubules against the flow of dentin fluid to the site of action, i.e. the nerve endings. Further, to induce depolarization of the nerves and achieve significant pain relief, the concentration of potassium must be sufficient surrounding the nerve ending, which may take 4 to 8 weeks. As the potassium level at the site of action is decreased, the relief of sensitivity is compromised.

c. *Oxalates*

The oxalate ions reacting with calcium ions in the dentinal fluid forming insoluble calcium oxalate crystals help in occluding dentinal tubules. However, these are acid labile and can be easily washed from the surface of dentin.

d. *Bioactive Glasses*

Bioactive glasses mainly contain silica along with calcium, sodium and phosphorus. Silica acts as nucleation site for precipitation of calcium and phosphate (crystalline hydroxycarbonate apatite), a product similar to dental apatite. This apatite has the potential to occlude dentinal tubules. Bioglass-based dentifrices are effective, being active for longer duration and are also acid resistant.

e. *Fluoride Salts*

It has been suggested that fluoride may mechanically block the tubules, or that labile fluoride in the organic matrix of the dentin could block the transmission of stimuli. Sodium fluoride is superior to sodium monofluorophosphate as regard fluoride deposition on the teeth.

Various other fluoride salts have been compared and evaluated; shown their efficacy as desensitizing agents.

f. *Milk*

A couple of studies have confirmed the efficacy of commercial milk as a desensitizing agent, especially after scaling and root planing. Milk is easily available, cheap and acceptable agent to be used as 'at-home' desensitizing agent.

g. *Hydroxyapatite Nanocrystals*

Nanotechnology implies processing of separation and consolidation of material by one atom or molecule (size of particles is roughly 2–3 atoms). The toothpaste containing carbonated nanohydroxyapatite crystals (nanotechnology) has been successfully used to reduce dentinal hypersensitivity. Hydroxyapatite crystals bind to enamel and dentin, producing a biomimetic coating on enamel/dentin. This coating prevents tooth from decay, helps revitalize the tooth and seals open dentinal tubules (treating dentinal hypersensitivity).

B. 'In-office' Treatment Modalities

The treatment modalities should primarily reduce the functional diameter of the tubules so as to limit fluid movement. In order to accomplish this objective, the possible approaches are:

i. Formation of a smear layer by brushing the exposed root surface.
ii. Topical application of agents that form insoluble precipitates within the tubules.
iii. Impregnation of tubules with plastic resins.
iv. Application of dental bonding agents to seal off the tubules.

Most of the desensitizing agents have the potential to occlude dentinal tubules; however, potassium nitrate acts by lowering the nerve sensitivity (Table 29.2).

Prior to treating sensitive root surfaces, hard or soft deposits should be removed from the teeth. Root planing may cause considerable discomfort; concerned teeth should be anesthetized prior to treatment. The teeth should be isolated and dried with warm air. When using desensitizing agents that have a caustic effect on soft tissue, care must be exercised to prevent them from contacting the alveolar mucosa.

In-office treatment modalities include:

a. Cavity varnishes
b. Corticosteroids

Table 29.2: Dentin desensitizers and mode of action

Chemical used (brand name)	*Method of evaluation*	*Mode of action*
Oxalate products (Sensodyne sealant, MS-coat)	• Air blast, tactile	• Tubule occlusion
Calcium phosphate products (Quell desensitizer)	• Quantitative permeability reduction	• Tubule occlusion
HEMA + Glutaraldehyde (Gluma desensitizer, Hema desensitizer, HemaSeal G)	• Tactile, air blast • Quantitative permeability reduction	• Protein precipitation/ tubule occlusion
Fluorides of sodium and stannous (Dentin bloc)	• Air blast, tactile	• Tubule occlusion
Fluoride varnish (5% NaF) (Durphat, cavity shield)	• Tactile, cold, air blast	• Tubule occlusion
Light-cure adhesives (Gluma, Prime and Bond 2.1)	• Air blast, tactile, cold • Quantitative permeability reduction	• Tubule occlusion
Glass-ionomer cement (Gluma Fuji VII)	• Tactile, cold, air blast	• Tubule occlusion
Potassium-containing products (potassium nitrate alone and in combination with other salts) (Radent, relief)	• Air blast, cold, tactile	• Lower nerve sensitivity

c. Procedures/chemical agents that can obturate dentinal tubules
   i. Burnishing of dentin
   ii. Silver nitrate/zinc chloride/strontium chloride/potassium oxalate
   iii. Calcium hydroxide/calcium phosphate
   iv. Sodium fluoride/sodium silicofluoride/stannous fluoride
d. Primers/bonding agents
e. Iontophoresis
f. Lasers
g. Restorative materials

a. Cavity Varnishes

Dentin often becomes insensitive when open tubules are covered with a thin film of varnish. This may be an effective means of providing temporary relief.

A fluoride-containing varnish (Duraflor) can be used for sustained relief.

It has been established that the effectiveness of varnish is limited to the duration for which it remained on the tooth surface.

b. Corticosteroids

Steroids have been widely used to treat dentinal hypersensitivity. It is reported that steroids cause complete obliteration of tubules thus decreasing dentin permeability.

c. Procedures/Chemical Agents that can Obturate Dentinal Tubules

i. ***Burnishing of dentin:*** Burnishing of dentin results in formation of a smear layer that partially occlude the dentinal tubules. Usually, wooden sticks are used to burnish dentin. It is observed that burnishing creates a partial smear layer that reduces fluid movement across dentin by 50 to 80%.

ii. ***Silver nitrate/zinc chloride/strontium chloride/potassium oxalate:*** Certain salts react with tooth structure to form crystals on the surface of dentin. The crystallization may occur within 1 to 2 minutes and such crystals can easily enter the dentinal tubules.

Silver nitrate precipitates protein constituents of odontoblast processes, thereby partially blocking the tubules.

Zinc chloride alone or zinc chloride–potassium ferrocyanide combination has been successfully used (40% solution of aqueous zinc chloride followed by 20% solution of aqueous potassium ferrocyanide).

iii. ***Calcium hydroxide/calcium phosphate:*** Calcium hydroxide is being routinely used for the treatment of dentin hypersensitivity. The exact mechanism of action is unknown; however, it is hypothesized that it may block dentinal tubules or promote peritubular dentin formation.

Increasing the concentration of calcium ions around nerve fibers may result in decreased nerve excitability; therefore, calcium hydroxide might be capable of suppressing nerve activity.

Calcium phosphate has also been successfully tried as a desensitizing agent.

iv. ***Sodium fluoride/sodium silicofluoride/stannous fluoride:*** 2.0% sodium fluoride has been routinely used as a desensitizing agent.

The dentinal fluid is saturated with calcium and phosphate ions and the application of sodium fluoride to dentin leads to precipitation of calcium fluoride crystals, thus reducing the functional radius of the dentinal tubules. The crystal size of calcium fluoride is very small (approximately 0.5 μm) and, therefore, less effective as compared to potassium oxalate that gives rise to larger crystals. Furthermore, it has been shown that fluoride is lost rapidly following application of sodium fluoride to dentin. Sodium silicofluoride is more effective than sodium fluoride. 10% solution of stannous fluoride has also been successfully used.

d. *Primers/Bonding Agents*

Various primers/bonding agents are used to seal the dentinal tubules.

The area of sensitive dentin is cleaned and etched with an acid conditioner for 5 seconds. The dentin is dehydrated with a continuous blast of air for at least 15 to 20 seconds in order to dry the outer part of the dentinal tubules. A drop of a bonding agent is applied to the dentin and cured.

A few bonding agents (Gluma desensitizer) contain glutaraldehyde and fluorides. Glutaraldehyde causes coagulation of the proteins inside the dentinal tubules, causing its precipitation and occluding the tubules.

e. *Iontophoresis*

Iontophoresis implies use of an electrical potential to transfer ions into the target object.

When a direct current passes through an electrolyte, positive ions travel toward the negative electrode (cathode) and negative ions towards the positive electrode (anode).

Iontophoresis involves the placement of a negative electrode to dentin and a positive electrode to the patient's face or arm. It is recommended that teeth should be isolated with plastic strips or cotton rolls, rather than with rubber dam. The moisture that accumulates between the tooth and the rubber dam may provide a low resistance pathway for the current.

The chemical is applied to the tooth surface and the current (0.5 mA) is passed through negative electrode.

Potassium oxalate (30%) and ferric oxalate (6%) solutions make available oxalate ions that react with calcium ions in the dentinal fluid to form insoluble calcium oxalate crystals. These crystals block the dentinal tubules.

It is established that application of potassium oxalate to the etched dentin also reduces sensory nerve excitability.

f. *Lasers*

Lasers have provided effective treatment option for dentin hypersensitivity.

Various types of lasers used are:

- CO_2 laser (10600 nm)
- Nd:YAG laser (1064 nm)
- Er:YAG laser (2940 nm)
- He:Ne laser (632.5 nm)
- Ga AlAs laser (810 nm)

It is demonstrated that when laser light acts on cell membrane, it allows greater passage and increase in calcium, sodium and potassium ions. Consequently, the endorphin system of neural cells increase and depolarization of C-fibers is blocked (do not allow pain information to reach central nervous system).

Most lasers occlude the dentinal tubules by producing local changes around the exposed dentin. Ga AlAs laser act by affecting the neural transmission in the dentinal tubules. It is hypothesized that lasers coagulate the proteins inside the dentinal tubules and block the movement of fluid. It is established that lasers are effective in reducing dentinal hypersensitivity without damage to pulp or any other adverse effect (Temperature changes at the level of pulp are generally less than 1.5°C with use of lasers; rise in temperature above 5.0°C affects pulp).

The limitations of using laser are being expensive and not easily available for routine clinical use.

g. *Restorative Materials*

Glass-ionomer luting cement has been successfully used to desensitize the dentin because it is hydrophilic, adheres well, and is esthetically pleasing.

The exposed dentin is restored with glass-ionomer cement/composite resins, especially in case of recession. A routine conservative cavity is prepared and restored with any of these materials.

NEWER MATERIALS

1. Arginine

It is established that saliva transports calcium and phosphate into dentin tubules inducing tubule plugging. A new 'saliva-based composition' containing arginine (an amino acid which is positively charged

at physiological pH), bicarbonate (a pH buffer), and calcium carbonate (a source of calcium) is developed to manage sensitivity. ProClude is one commercial preparation. Clinical studies have shown that this paste is effective in providing instant sensitivity relief and that the relief lasts for at least 28 days.

Colgate has further improved this combination by combining arginine, calcium carbonate and fluoride to provide a significant advancement. A dentifrice containing 8.0% arginine, calcium carbonate, and 1450 ppm fluoride, has been clinically proven to provide lasting relief to sensitivity.

Confocal laser scanning microscopy (CLSM) has confirmed that arginine effectively plugs and seals dentin tubules, more so the occlusion achieved is resistant to acid challenge. CLSM studies have also shown that arginine is delivered to the inner surfaces of dentin tubules within the occluded dentin plug. The arginine-containing toothpaste results in reduced dentin fluid flow, inhibiting the hydrodynamic mechanism.

Arginine is clinically proven to provide instant sensitivity relief prior to and after dental procedures. Arginine-containing toothpaste was effective when compared to potassium-salt toothpaste and low in comparison to strontium-containing toothpastes. However, effectiveness of arginine is low in mouthwashes (Professionally applied arginine showed better results).

2. NovaMin

NovaMin is bioactive glass ceramic material that provides calcium and phosphate. The active ingredient is calcium sodium phosphosilicate that reacts when exposed to aqueous medium and provides calcium and phosphate ions that form hydroxy-carbonate apatite with time.

NovaMin is successfully used to treat dentinal hypersensitivity. The calcium and phosphate ions block the dentinal tubules, thereby reducing the flow of dentinal fluid. NovaMin has been used with and without fluoride (Oravive). NovaMin in combination with 5% sodium fluoride (Renew: 5 wt% NovaMin and 5000 ppm fluoride) is also marketed. The manufacturers also claim its ability to whiten and remineralize the affected dentin.

3. Casein Derivatives

Casein is the predominant phosphoprotein in bovine milk and accounts for almost 80% of its total protein. Under alkaline conditions, the calcium phosphate is present as an alkaline amorphous phase along with calcium phosphate, referred to as casein phosphopeptide-amorphous calcium phosphate (CPP-ACP). They have the ability to stabilize high concentrations of calcium and phosphate in metastable solutions. CPP-ACP is a useful cariostatic agent as it contains calcium and phosphate in a bio-available form. CPP-ACP has been developed as a remineralizing agent (GC Tooth Mousse); can remineralize enamel subsurface lesions. Because of its remineralizing capacity, it is also recommended as desensitizing agent.

CPP-ACP can be used as an:

- Adjunct preventive therapy to reduce caries in high-risk patients
- Reduce dental erosion in patients with gastric reflux
- To repair enamel involving white spot lesions
- Repair orthodontic decalcifications
- Desensitizing agent

The mode of action of certain desensitizing agents is summarized in Table 29.3.

A few studies evaluating treatment modalities for dentin hypersensitivity are summarized in Table 29.4.

PREVENTION

Suggestions for Patients

- Avoid gingival recession by practicing good oral hygiene measures.
- Avoid using large amounts of dentifrices.

Table 29.3: Dentin hypersensitivity: Chemical agents and mechanisms of action

Chemical agent	*Mechanism of action*
Pro-arginine	Occlusion of tubules by precipitation of glycoprotein
Novamin	Occlusion of tubules by precipitation of calcium and phosphate
Potassium salts	Blocking the neural response
Strontium/stannous salts	Occlusion of dentin tubules by the precipitation of metallic ions
Oxalates	Seal the tubules by calcium oxalates
Adhesive agents	Seal the tubules forming hybrid layer
Laser	Depolarization of nerve fibers and occlusion of tubules
Root coverage	Coverage of exposed dentin with surrounding gingiva

Table 29.4: Treatment modalities for dentin hypersensitivity: Studies evaluated

Author (year)	*Sample size and study design*	*Time of evaluation (method of assessment)*	*Results*
Arginine and calcium carbonate			
Chu and Lo (2014)	Randomized study: 303 samples (41% males, 59% females)	Immediate (airblast)	8.0% Arginine–calcium carbonate paste, when used immediately on sensitive teeth, significantly improved sensitivity.
Hu, et al (2013)	Randomized study: 90 samples (arginine vs control)	2, 4, 8 weeks (tactile, airblast)	Arginine mouthwash exhibited statistically significant improvement in hypersensitivity scores as compared to the Control Mouthwash.
Sowinski, et al (2013)	Randomized study: 100 samples (arginine vs fluorides)	8 weeks (tactile, airblast)	Arginine group demonstrated improvements in tactile/airblast sensitivity as compared to fluoride dentifrices.
Tsai, et al (2012)	Randomized study: 66 samples (arginine vs control paste)	Immediate (airblast)	Significant decrease (83.94%) in the hypersensitivity of subjects using arginine paste.
Polymers			
Chaknis, et al (2011)	Double-blind, randomized study: 118 samples Dentifrice 1(0.3% triclosan, 2.0% copolymer, 0.2% sodium fluoride) Dentifrice 2 (0.4% stannous fluoride, zinc lactate) Dentifrice 3 (0.245 sodium fluoride)	8 weeks (tactile, airblast)	Significant decrease in sensitivity reported in all the three groups; especially at 8 weeks observation.
Potassium salts			
Veitz-Keenan, et al (2013)	Randomized study: 293 samples (sealant/restoration vs dentifrice)	1, 3, 6 months (airblast)	The sealant and restoration groups observed a significantly higher reduction in sensitivity than the dentifrice group.
Liu and Hu (2012) difference	Double-blind, randomized study: 81 samples (topical use: 40 vs control: 41)	Immediate, after 3 days (tactile, airblast)	Topical use (experimental) observed significantly decrease in sensitivity after 3 days; not much immediately. Both tactile and airblast methods significantly observed the same results.
Wara-Aswapati, et al (2005)	Double-blind, randomized study: 102 samples (paste containing potassium nitrate and triclosan vs control)	4, 12 weeks (tactile, airblast)	Significantly effective as compared to control group
Casein derivatives (amorphous calcium phosphate (ACP) and casein phosphopeptide (CPP)			
Geiger, et al (2003)	Randomized study: 30 samples (experimental vs placebo)	7, 28, 180 days (tactile, airblast)	Both the experimental and placebo resulted in a reduction in hypersensitivity at 180 days; however, the experimental group showed a rapid reduction in hypersensitivity.
Stannous fluoride			
He, et al (2014)	Double-blind, randomized study: 116 samples (stannous fluoride dentifrice vs control)	Immediate, 3 days, 2 weeks (tactile, airblast)	The stannous fluoride dentifrice was superior as compared to the control dentifrice (the timings and method of evaluation correlated with the observation).

Contd...

Table 29.4: Treatment modalities for dentin hypersensitivity: studies evaluated (*Contd...*)

Author (year)	*Sample size and study design*	*Time of evaluation (method of assessment)*	*Results*
Ni, et al (2010)	Double-blind, randomized study: 58 samples (stannous fluoride dentifrice vs control)	4, 8 weeks (tactile, airblast)	Both the experimental (stannous fluoride containing) and control dentifrices provided significant reductions in hypersensitivity.
Calcium sodium phosphosilicate (CSPS)			
Sharma, et al (2010)	Double-blind, randomized study: 120 samples (calcium sodium phosphosilicate vs stannous fluoride vs potassium nitrate)	2, 4, 12 weeks (tactile, airblast)	All three groups significantly reduced sensitivity at each time interval; calcium sodium phosphosilicate (NovaMin®) dentifrice reduced sensitivity significantly more than the others at the 2- and 4-week time interval.
Acharya, et al (2013)	Double-blind, randomized study: 20 samples (calcium sodium phospho-silicate vs potassium nitrate)	2, 4, 8 weeks (airblast)	The calcium sodium phosphosilicate group showed a higher degree of reduction of hypersensitivity than potassium nitrate after 2 weeks; however, no significant difference was observed at 4 weeks and 8 weeks.
Oxalates			
Craig, et al (2012)	Double-blind, randomized study: 10 samples (silver fluoride/potassium iodide vs oxalates)	1 week (thermal)	Silver fluoride/potassium iodide group was significantly better than the oxalate group.
Resin-based materials			
Sethna, et al (2011)	Randomized study: 250 samples (Cervitec vs Gluma varnish)	Immediate, 4, 12 weeks (tactile, airblast, thermal)	Both varnishes were effective in alleviating dentin hypersensitivity at all times; however, Cervitec varnish was more effective than Gluma varnish at both 4 weeks and 12 weeks.
Ding, et al (2014)	Double-blind, randomized study: 119 samples (Gluma varnish vs control)	Immediate, 1, 4 weeks (tactile, airblast)	Gluma varnish was significantly superior to control group at all times
Lasers			
Flecha, et al (2013)	Double-blind, randomized study: 434 samples (laser, cyanoacrylate vs control)	24 hours, 30, 90, 180 days (airblast, thermal)	Laser cyanoacrylate were superior to control at all times.

- Avoid toothbrushes with hard bristles.
- Avoid brushing teeth immediately following ingestion of acidic food or beverages.
- Avoid over brushing with excessive pressure for prolonged periods of time.
- Avoid excessive flossing or incorrect use of other interproximal cleaning devices.
- Avoid frequent use of toothpicks.

Suggestions for Professionals

- Avoid over instrumenting the root surfaces during calculus removal and scaling and root planing.
- Avoid over polishing the exposed roots during stain removal.
- Avoid violating the biologic width when placing crown margins causing subsequent recession.
- Avoid 'burning' the gingival tissue during in-office tooth whitening or bleaching procedures.

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